Particulars of the Experiments Performed

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Introduction to MATLAB.

What is MATLAB ?

MATLAB stands for MATrix LABoratory. It is a high-level programming language & numerical computing software developed by Math works. Math is widely used in engineering, Science, mathematics, & research for solving complex numerical problems, data analysis, visualization & simulation

Why MATLAB ?

Mechanical engineers often encounter problems involving.

- Matix calculations
- Numerical methods.
- differential equation
- optimization
- Data Analysis.
- Control System.

MATLAB provides a simple & different efficient environment to solve these engineering problems.

Features of MATLAB.

- Easy-to-learn programming language.
- powerful matrix manipulation capabilities
- Extensive mathematical function.
- High-quality graphical visualization.

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2. MATLAB Environment :-

When MATLAB is opened. the following windows are available.

1. Command Window.
used to enter commands & execute them directly

Example :-

```
MATLAB
>> 5+10
MYJ22ME72_SHD
MATLAB BASICS
ans = 15
```

2. Workspace.
Stores all variables currently being used.

Example :

MATLAB.

```
a = 10;
b = 20;
```

Variable a and b appear in the workspace.

3. Current folder

Displays files & scripts stored in the working directory.

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4. Editor Window
used for writing & editing MATLAB programs (.m files).

5. Command History.
Stores previously executed commands.

3. Advantages & disadvantages of MATLAB.

Advantages :-

- Simple & user-friendly.
- Fast numerical computation.
- Excellent plotting & visualization tools.

disadvantages :-

- Expensive licensed software.
- Requires considerable memory for large problems.
- slower than low-level languages like C & C++.
- proprietary software.

4. Applications :-

- Mechanical Engineering.
- Vibration analysis
- Machine design
- Heat transfer
- Fluid mechanics.

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- Electrical Engineering
- civil Engineering.
- Aerospace Engineering.

5. MATLAB BASICS

Variable :- used to store data.

Ex :- MATLAB
```
a = 10;
b = 20;
c = a+b;
output c = 30.
```

Rules for Naming variables.
- must start with a letter
- can contain letters, numbers & underscores
- cannot contain space
- MATLAB is case-sensitive.

Data Types.

Numeric
```
a = 25;
```

character
```
name = 'MATLAB';
```

Logical
```
x = true;
```

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Exp. Result ..... Exp No. ..... Page No. 3

6 Arithmetic operations :-

operation	MATLAB Symbol	Example
Addition	+	a+b
Subtraction	-	a-b
Multiplication	*	a*b
division	/	a/b
power	^	a^2

Built-in Mathematical functions

Square root
```
sqrt(25)
output 5
```

Exponential
```
exp(2)
```

Logarithm
```
log(10)
```

Trigonometric function
```
sin(pi/2)
cos(pi)
tan(pi/4)
```

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7. Matrices & Arrays

MATLAB is a matrix-based software.

Creating a Matrix.

A = [1,2,3;4,5,6];

3
7,8,9];

Accessing Elements

A(2,3)
output:
6

8. Scripts & functions :-

Script file.

A script is a collection of MATLAB commands saved in a file with extension .m.

Example.

```
clc;
clear;
a=10;
b=20;
c= a+b
```

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Exp. Result ..... Exp. No. .....

Save as : addition.m

```
clc
```

clear command window

```
clear
```

Clear Workspace variables
close all

Function file :- performs a specific task & returns output.

9. Control statements

If statement.

```
a = 10
if a>0
disp ('positive')
```

For loop.

While loop.

```
i=1;
while i<=5
disp (i)
5
i = i+1
```

6
end.

10. plotting Graphs in MATLAB.

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Exp. Result ..... Exp. No. .....

6
end.

10 plotting Graphs :-

Simple plot

```
x = 0:0.1:10;
y = sin(x);
plot(x,y)
```

Adding labels.

```
xlabel('Time')
ylabel('Amplitude')
title('Sine Wave')
grid on.
```

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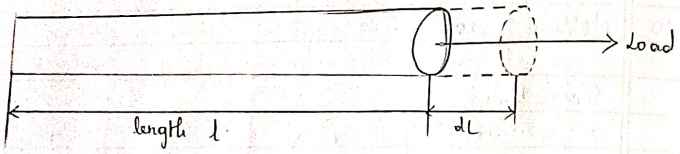

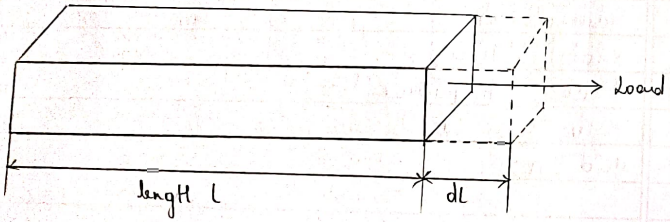

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Exp. Result..... Exp. No. 01

Exp no:-1

stress & strain for a Bar subjected to axial load.

Aim:- To calculate stress & strain for a bar subjected to axial load using MATLAB program.

Theory:- A bar subjected to an axial load undergoes deformation due to tension (or) compression. The resistance offered by the material against deformation is called stress, while the ratio of change in length to orginal length is called strain.

According to Hooke's law, within the elastic limit stress is directly proportional to strain:

$$\sigma = P/A$$

$$\varepsilon = dl/l.$$

Where σ = stress (MPa)

P = Applied load (N)

A = cross-sectional area (mm^2)

ε = strain

dl = change in length (mm)

l = original length (mm)

The relationship between stress & strain is represented by young's modulus (E):- $E = \frac{\sigma}{\varepsilon}$

Tabular column :-

Sl No	Load (P) Newton	Stress σ MPa	Elongation dL mm	Strain ε
1	1000	4	0.004	2×10^{-5}
2	2000	8	0.008	4×10^{-5}
3	3000	12	0.012	6×10^{-5}
4	4000	16	0.016	8×10^{-5}
5	5000	20	0.02	1×10^{-4}
6	6000	24	0.024	1.2×10^{-4}
7	7000	28	0.028	1.4×10^{-4}
8	8000	32	0.032	1.6×10^{-4}
9	9000	36	0.036	1.8×10^{-4}
10	10000	40	0.04	2×10^{-4}

Calculations :-

$E = \frac{\sigma}{\varepsilon} = \frac{1000 \times 2000}{250 \times 2 \times 10^5} = 0.04$ mm $\sigma = \frac{P}{A} = \frac{1000}{250} = 4$ MPa.

$\varepsilon = \Delta L / L = 0.04 / 2000 = 2 \times 10^{-5}$.

P = 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000

Stress = 4 8 12 16 20 24 28 32 36 40

Elongation = 0.04, 0.08, 0.12, 0.16, 0.20, 0.24, 0.28, 0.32, 0.36, 0.40

Strain = 2×10^{-5}, 4×10^{-5}, 6×10^{-5}, 8×10^{-5}, 10×10^{-5}, 12×10^{-5}, 14×10^{-5}, 16×10^{-5}, 18×10^{-5}, 20×10^{-5}.

Name of the Experiment ..... Date 13/8/25 Page No. 7

Exp. Result ..... Exp. No. 01

Tools Required :- Command Window in MATLAB

Procedure :-

A = 250;

L = 2000;

E = 2e5

P = [1000 2000 3000 4000 5000 6000 7000 8000 9000 10000];

Stress = P/A

Elongation = (P*L)/(A*E)

Strain = Elongation/L

Tabulation :-

The area of the bar A = 250 mm²

The length of bar L = 2000 mm

Young's modulus E = 2e5 MPa.

Conclusion :- The results from MATLAB program & manual calculations are same.

Name of the Experiment..... Date 20/8/26

Exp. Result ..... Exp. No. 02 Page No. 8

Exp no :- 2

Deflection & potential Energy of spring subjected to different load steps

Aim :- To write a script to plot the deflection & potential energy of spring subjected to different load steps in MATLAB

Theory :- A spring is an elastic element that store mechanical energy when it is compressed or extended. The force developed by a spring is approximately proportional to its deflection according to Hooke's law.

The spring stiffness is given by

$$k = F/x$$

When a spring is loaded, it deforms its original position & store potential energy due to deformation. The stiffness of springs depends on their arrangement. For springs in parallel, stiffness is additive.

Where K = Spring stiffness (N/mm)
F = Applied force (N)
x = Extension / Compression (mm)

The potential energy stored is

$$U = \left(\frac{1}{2}\right) kx^2$$

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Tabular column :-

Sl No	Load (F) in Newton	deformation x in mm	potential energy stored in N-mm²
1	40	2	40
2	50	2.5	62.5
3	60	3	90
4	70	3.5	122.5
5	80	4	160

Calculations :-

$x = \frac{F}{k} = \frac{40}{20} = 2\,mm$

$u = \left(\frac{1}{2}\right) kx^2 = \left(\frac{1}{2}\right) \times 20 \times 2^2 = 40\,N/mm^2$.

⇒ MATLAB :-

F = (N) = 40, 50, 60, 70, 80.

x = (mm) = 2, 2.5, 3, 3.5, 4.

u = (N/mm²) = 40, 62.5, 90, 122.5, 160.

Name of the Experiment..... Date 20/8/26 Page No. 9

Exp. Result ..... Exp. No. 02.

where u = potential energy stored in N-mm²

Tools Required :- Command Window, Script in MATLAB

Procedure :-

```
k = 20;
F = [40 50 60 70 80];
x = F/k
u = (1/2)*k*x.^2.
```

Tabulation :-

The spring rate (spring stiffness) k = 20 N/mm

Conclusion :- The results from MATLAB program & Manual calculations are same.

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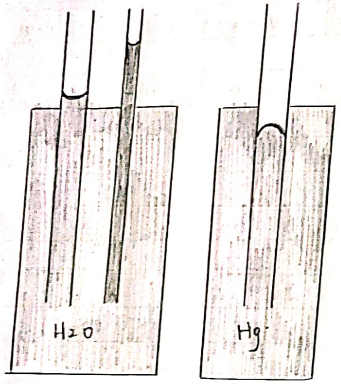

Name of the Experiment..... Date 27/8/26 Page No. 10

Exp. Result ..... Exp. No. 03

Exp no:- 3

MATLAB function to calculate the capillary rise of liquid in A tube.

Aim :- To write a MATLAB function to calculate the capilary rise of liquid in a tube.

Theory :- Capillary action is the movement of liquid through a narrow tube (or) space due to adhesive & cohesive forces, even against gravity. It is commonly observed in thin tubes & porous materials.

The rise (or) fall of liquid level in a thin tube is called capillary rise or capillary fall. The capillary height decreases as the tube dia increases.

For a liquid in a narrow tube surface tension & adhesive forces between the liquid & tube wall cause the liquid to rise (or) fall.

The capillary rise / fall is given by :-

$$h = \frac{2T\cos\theta}{r\rho g}$$

where h = capillary rise / fall in m

T = surface tension on liquid (72.7×10^{-3} N/m for water)

r = radius of tube in m.

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Tabular column :-

Sl No	Angle of contact (θ in degrees)	Radius to tube r	Capillary rise h
1	0	10mm	1.48×10^{-3}
2	15		1.43×10^{-3}
3	30		1.28×10^{-3}
4	45		1.048×10^{-3}
5	0	20mm.	7.33×10^{-4}
6	15		7.15×10^{-4}
7	30		6.41×10^{-4}
8	45		5.24×10^{-4}.

Calculations :-

$r = \frac{10}{1000} = 10 \times 10^{-3}$ m $\quad \frac{20}{1000} = 20 \times 10^{-3}$ m.

$h = \frac{2T\cos\theta}{\rho g} = \frac{2 \times 72.7 \times 10^{-3} \times \cos 0}{10 \times 10^{-3} \times 1000 \times 9.81} = 1.48 \times 10^{-3}$ m.

$h = \frac{2T\cos\theta}{\rho g} = \frac{2 \times 72.7 \times 10^{-3} \times \cos 0}{20 \times 10^{-3} \times 1000 \times 9.81} = 7.33 \times 10^{-4}$

Name of the Experiment..... Date 27/8/26 Page No. 11

Exp. Result ..... Exp. No. 03

θ = Angle of contact between liquid surface & the tube (0 degree for tube vertical & 90 degree if tube is horizontal).

ρ = density of liquid (1000 kg/m^3)

g = Acceleration due to gravity.

Tools Required :- Command Window, Script in MATLAB.

Procedure :-

```
function [h] = caprise (theta, T, r)
theta = [0 15 30 45];
r = 0.01;
T = 72.7 * 10^-3;
rho = 1000;
g = 9.81;
h = (2* T* cosd (theta))/(r*rho *g);
```

Conclusion :- The results from MATLAB program & manual calculations are same.

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Name of the Experiment..... Date ..... Page No. 12

Exp. Result ..... Exp. No. .....

Exp no:-4.

MATLAB program to find the pressure & velocity of a liquid in pipe using Bernoullis equation by solving a system of linear equations.

Aim:- To write a MATLAB program to find the pressure & velocity of a liquid in pipe using Bernoullis equation by solving a system of linear equations.

Theory:- Bernoullis principle states that in a steady flow of an ideal fluid, the total energy per unit weight remain constant along a stream-line. An increase in fluid velocity results in a decrease in static pressure, provided the elevation remain constant.

Bernoullis equation is given by.

$$\frac{P}{\rho g} + \frac{v^2}{2g} + z = \text{constant}.$$

where P = Pressure of liquid (Pa or N/m^2)
ρ = density of liquid (for water 1000 kg/m^3)
g = Acceleration due to gravity (9.81 (m/s^2))
v = velocity of flow in m/s.
z = datum head in m.

Calculation

$$\frac{P_1}{1000\times9.81} + \frac{1}{2\times9.81} + 4 = \frac{1}{1000\times9.81} + \frac{1/2}{2\times9.81} + 10.$$

$$H_2 - H_1 = \frac{V_2^2 - V_1^2}{2g} + (Z_2 - Z_1)$$

$$21.4679 - 19.2905 = \frac{0.5V_1^2 - (1)^2}{2\times9.81} + (10-4)$$

$$2.1774 = \frac{-0.75V_1^2}{19.62} + 6.$$

$$2.1774 = \frac{-0.75V_1^2}{19.62} + 117.72$$

$$2.1774 \times 19.62 = -0.75V_1^2 + 117.72$$

$$42.7205 = -0.75V_1^2 + 117.72$$

$$42.7205 - 117.72 = -0.75V_1^2$$

$$-74.99 = -0.75V_1^2$$

$$V_1^2 = \frac{74.99}{0.75} = 100$$

$$V_1^2 = 100$$

$$V_1 = \sqrt{100} = 10\ m/s.$$

Name of the Experiment..... Date ..... Page No. 13

Exp. Result ..... Exp. No. .....

Procedure :-

A = [1/(1000 * 9.81), 1(2*9.81); 1/(1000 * 9.81)
1/(8 * 9.81)];

B = [15.2905; 11.47679];

X = A\B

pressure = x(1,1)

velocity = sqrt(x(2,1)).

Tabulation :-

* let pressure P be constant throughout the pipe.
* velocity at section 1 is v.
* velocity at section 2 is half of velocity at section 1
* section 1 is at a height of 4m & section 2 is at height of 10m.
* The total energy head at section 1 is 19.2905 m & at section 2 is 21.4679m.

$V_2 = \frac{0.5 \times 10}{V_2 \times V_1} = 5\ m/s$

$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = H \Rightarrow \frac{P_1}{1000 \times 9.81} + \frac{10^2}{2 \times 9.81} + 4 = 19.2905$

$\frac{P_1}{9810} + \frac{100}{19.62} + 4 = 19.2905$

$\frac{P_1}{9810} + 5.0968 + 4 = 19.2905$

$\frac{P_1}{9810} + 9.0968 = 19.2905$

$\frac{P_1}{9810} = 19.2905 - 9.0968$

$\frac{P_1}{9810} = 10.1936 = 10.1936 \times 9810$

$P_1 = 99999.805 \approx 1,00,000$

$P = 100\ kpa$

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Exp. Result ..... Exp. No. .....

Result :- The Matlab program result & manual calculations are same.

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Name of the Experiment..... Date ..... Page No. 15

Exp. Result ..... Exp. No. .....

Exp no :- 5.

To develop a model of the effect of temperature on viscosity by the use of curve fitting in MATLAB.

Aim :- To develop a model of the effect of temperature on viscosity by use of curve fitting in MATLAB.

Theory :- Viscosity is the measure of a fluids resistance to flow. liquids with viscosity like syrup flow slowly, while liquids with low viscosity like water flow easily. Viscosity is caused by friction between adjacent layers of fluid.

The viscosity of liquid of a temperature. Generally the viscosity of liquids decreases as temperature increases.

Andrade equation for the effect of temperature of viscosity is:

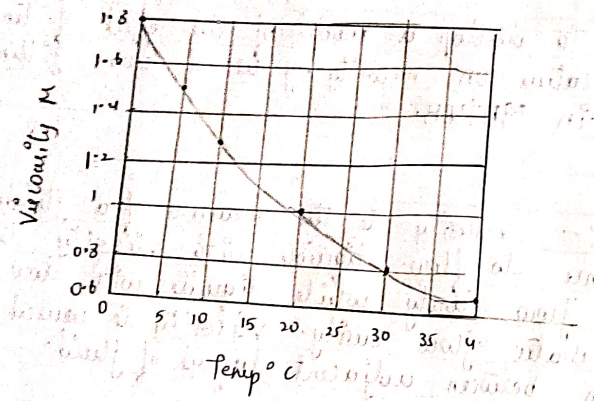
$$\mu = De^{B/Ta}.$$

Where :

μ = dynamic viscosity of water. (10^{-3} Ns/m^2)

Ta = Absolute temperature in K.

D & B are parameters.



Name of the Experiment ..... Date .....

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Procedure :-

```
% Experimental data
T = [0 5 10 20 30 40]; % Temperature in deg.
mu = [1.787 1.519 1.307 1.002 0.7975 0.6529];
% Convert temperature to Kelvin.
Ta = T + 273;
% Linearization of Andrade's Equation.
% mu = D * exp(B/Ta)
% ln(mu) = ln(D) + B*(1/Ta)
X = 1/Ta;
Y = log(mu);
% curve fitting using linear regression.
P = polyfit(X, Y, 1);
B = P(1);
lnD = P(2);
D = exp(lnD);
% predicted viscosity values.
mu_fit = D * exp(B./Ta);
% display results.
fprintf('Value of D = %f\n', D);
fprintf('Value of B = %f\n', B);
% plot experimental and fitted data figure;
plot(T, mu, 'ro', 'MarkerSize', 8, 'LineWidth', 2);
hold on;
plot(T, mu_fit, 'b-', 'LineWidth', 2);
xlabel('Temperature (°C)');
ylabel('Viscosity (\mu)');
```

MATLAB calculations :-

1.0e+03*
2.148 - 0.0073

Value of α = 0.000668

Value of B = 2148.544608

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Exp. Result ..... Exp. No. .....

```
title ('Effect of temperature on viscosity');
legend ('Experimental data', 'curve fit');
grid on;
```

Result :- The curve fitting tool box was successfully to obtain a model for given set of data

Conclusion :- The obtained curve best suits the test data

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Name of the Experiment..... Date ..... Page No. 18

Exp. Result ..... Exp. No. .....

Exp no : 6.

To develop A model of a steady flow of water in a circular pipe by the use of multiple linear regression.

Aim :- To develop a model of a steady flow of water in circular pipe by the use of multiple linear regression

Theory :- Linear regression is a statistical method used to find the relationship between dependent & independent variables. when one independent variable is used it is called simple linear regression & when 2 or more are used it is called multiple linear regression.

Tools Required :- Command Window, script in MATLAB.

Procedure :-

```
% Assumed experimental data
Q = [10, 20, 30, 40, 50, 60, 70];
P = [15, 32, 45, 61, 75, 83, 105];
% perform linear regression.
Co-efficients = polyfit (Q, P, 1)
m = Co-efficients (1);
c = Co-efficients (2);
% Generate the best fit line data
Q_fit = linspace (min (Q), max (Q), 100);
P_fit = polyval (coefficients, Q_fit);
```

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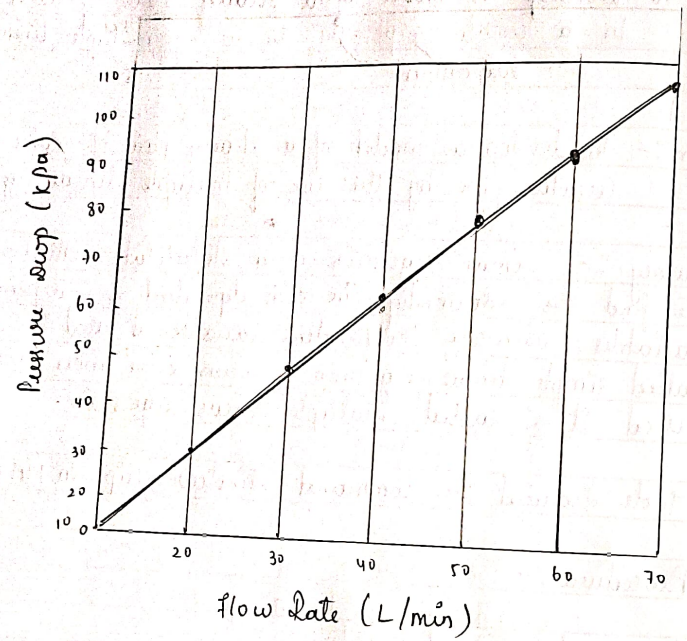

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Exp. Result ..... Exp. No. .....

```
figure;
plot (Q, P, 'ro', 'MarkerSize', 8, 'MarkerFaceColor', 'r');
hold on;
plot (Q_fit, P_fit, 'b-', 'LineWidth', 2);
title ('Linear Regression : Steady flow in a circular pipe')
X label ('Flow Rate (L/min)');
Y label ('Pressure Drop (kpa)');
legend ('Experimental data', 'Linear fit', 'Location', 'northwest');
grid on;
hold off;
```

Result :- The curve fitting tool box was successfully obtained a model for given set of data.

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Exp no: 7

Animation of displacement of slider crank mechanism

Aim :- To write a code to plot the animation of displacement of slider crank mechanism with respect to input angle

Theory :- The slider crank mechanism is a particular four-bar linkage configuration that converts linear motion to rotational or vice versa..

The displacement of slider of the slider-crank mechanism shown in fig is given by $s = a\cos(\phi) + \sqrt{b^2 - (a\sin(\phi) - e)^2}$

$a=1$, $b=1.5$ $e=0.3$

Tools Required :- Command prompt in MATLAB.

Procedure :-

```
% Simple simulation for the crank-slider mechanism.
% l = rod length (distance between piston pin and crankpin)
% r = crank radius (distance between crank pin and crank
      center, i.e. half stroke)
% A = crank angle
% x = piston pin position.
l = 30;
r = 15;
X = [0 0 0];
Y = [0 r r+l];
```

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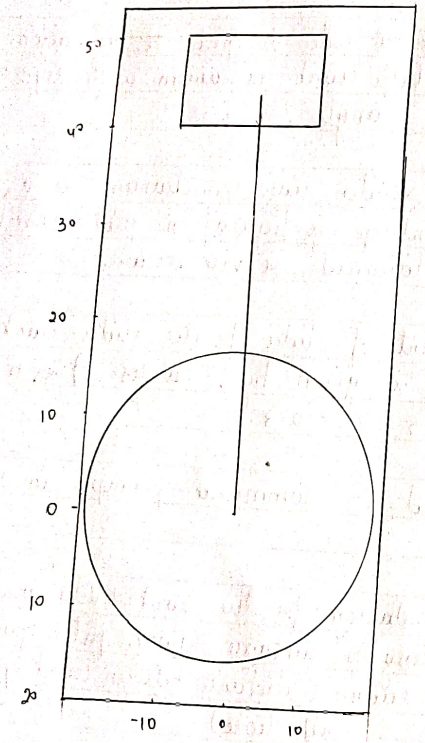

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Exp. Result ..... Exp. No. .....

```
X_Piston = [-7 7 7 -7 -7];
Y_piston = [1-5 1-5 1+5 1+5 1-5];
h = plot (X, Y, 'Line Width', 4, 'XData Source', 'X', 'YData Source', 'Y');
axis ([-1.1*r 1.1*r -1.5*r 1.5*r+1]);
set (gca, 'Data Aspect Ratio', [1 1 1])
grid on
hold ('all')
g =
plot (X_piston, Y_piston, 'r', LineWidth', 4, 'XData Source', 'X_piston',
    'YData Source', 'Y_piston');
angle = 0:0.01:2*pi;
x_circle = r.* cos (angle);
y_circle = r.* sin (angle);
i = plot (x_circle, y_circle, 'Line Width', 4)
for A = 0:0.08:4* pi
x = A* cos (A) + sqrt (12 - r12 * sin (A)^2);
N = [r* sin (A) r* cos (A)];
P = [0 x];
X = [0 r* sin (A) 0];
Y = [0 r* cos (A) x];
Y_Piston = [x-5 x-5 x+5 x+5 x-5];
refreshdata (h, 'caller')
refreshdata (g, 'caller')
refreshdata (i, 'caller')
drawnow
pause (.1)
end
```

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Exp. Result ..... Exp. No. .....

Result :- The movement of piston we got in animation the piston move from TDC - BDC

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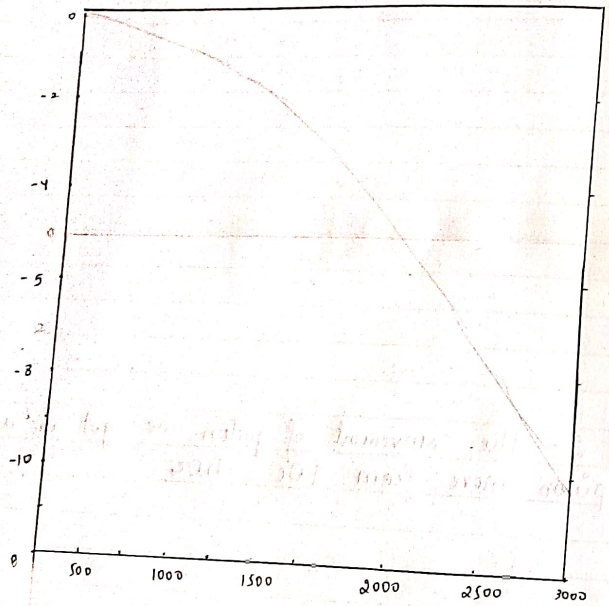

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Exp. Result ..... Exp. No. .....

Exp no: 8.

Deflection of beam

Aim :- To solve a differential equation to calculate the deflection of the beam.

Theory :- To determine the slope & deflection of the beam at any given point the following 2nd order linear differential equation which governs the elastic curve characterizing the shape of the deformed beam is used.

$$\frac{d^2y}{dx^2} = \frac{M(x)}{EI}.$$

The deflection equation for a cantilever beam with point load at the free end is

$$y = \frac{P}{6EI}\left(-x^3 + 3L^2x - 2L^3\right)$$

Tools Required :- Editor & command prompt in MATLAB.

Procedure :-

i Create a function file:

```
function [d2y] = odfun(x,y)
d2y = zeros(2,1);
```

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```
dy(1) = y(2);
dy(2) = -25000 * (3000-x)/(2.1e5) * 1e8);
end
```

Command Prompt:

```
[x,y] = ode45(@odfun, [0 3000], [0 0])
plot (x,y(:,1))
```

Result :- The given matlab code is executed successfully.

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Exp no : 9.

Minimum drag on an airfoil

Aim :- To calculate the minimum drag on a airfoil by the use of optimization technique

Theory :- An object in fluid flow experience a net aerodynamic force, which resolves into drag & lift relative to flow direction.

Drag (D) : Resistive force parallel to flow

Lift (L) : force normal to flow.

Total drag is given by

$$D = 0.01 \sigma V^2 + 0.95 \left(\frac{W}{V}\right)^2 \frac{1}{\sigma}$$

where σ = air density ratio

W = weight

V = velocity

Tools Required : Editor & command prompt in MATLAB.

Procedure :-

Create a function file :

function [drag] = aerofun(v)

Output

drag = 2.5000 e-04

Velocity = 145920000.

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```
sigma = 0.6;
W = 16000;
drag = (0.01 * sigma * v * v) + (0.95 * W * W / (sigma * v * v));
end
```

Command prompt

```
[drag, velocity] = fminsearch (@zerofun.0)
```

Result :- The given matlab code is executed successfully.

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